

# EE5750 Quantum Error Correction

## Homework 4

Due: 27 Aug 2026

As part of the Homework assessment, provide solutions to only the questions marked as **(\*Homework\*)**. The rest of the questions are practice problems.

1. **(\*Homework\*)** (5 marks) *Positive quadratic forms and Hermitian matrices.*

In the lectures, we defined an operator  $A$  to be positive, i.e.,  $A \geq 0$  when

- $A^\dagger = A$ , and
- $\langle \psi | A | \psi \rangle$  is real and non-negative for all  $|\psi\rangle$ .

We will now show that the second condition implies the first, and hence, the first condition is superfluous. Our task now is to show the following

$$\langle \psi | A | \psi \rangle \geq 0 \text{ for all } |\psi\rangle \Rightarrow A^\dagger = A.$$

To do so we will use the fact  $A = B + iC$  for some Hermitian operators  $B, C$  (see Homework 1).

- (a) Considering  $\langle \psi | A | \psi \rangle$ , which is real, what can we say about  $\langle \psi | B | \psi \rangle$  and  $\langle \psi | iC | \psi \rangle$ ? Are they both real? Is one of them equal to 0?
- (b)  $C$  is Hermitian and from part (a) we know that  $\langle \psi | C | \psi \rangle = 0$  for every  $|\psi\rangle$ . Show that all eigenvalues of  $C$  are zero, and using this show that  $C = 0$ . This implies that  $A = B$  is Hermitian.

Hint: Use the spectral decomposition of  $C$ , that is,  $C$  is unitarily diagonalizable

$$C = \sum_k \lambda_k |\phi_k\rangle \langle \phi_k|$$

where  $\{|\phi_k\rangle : k = 1, 2, \dots\}$  is an orthonormal basis of the associated Hilbert space.

Remark: If every eigenvalue of an operator is 0, it does not necessarily mean that the operator is 0, for example

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

In the question we are interested in,  $C$  is diagonalizable, and this helps us prove that  $C = 0$ .

2. **(\*Homework\*)** (5 marks) The *Hilbert-Schmidt* inner product between operators  $A$  and  $B$  is defined as

$$\langle A, B \rangle = \text{Tr}(A^\dagger B).$$

This is an inner product on the vector space of operators. Note that  $\langle A, B \rangle$  and  $\langle B, A \rangle$  are conjugates of each other. You must prove the following:

- (a) Show that  $\langle A, A \rangle \geq 0$ .

Hint: What kind of operator is  $A^\dagger A$ ?

- (b) (Remember the 4 Pauli matrices  $I, X, Y, Z$ ; it is good to know them by heart.) For any two distinct Paulis, verify that the Hilbert-Schmidt inner product is zero.
- (c) Recall that any  $2 \times 2$  matrix  $A$  can be written as

$$A = c_1 I + c_2 X + c_3 Y + c_4 Z$$

for some complex scalars  $c_1, \dots, c_4$ . Show that  $c_1 = \text{Tr}(A)/2$  and  $c_2 = \langle X, A \rangle/2$ . What about  $c_3$  and  $c_4$ ?

Hint:  $\text{Tr}(\alpha A + \beta B) = \alpha \text{Tr}(A) + \beta \text{Tr}(B)$ .

Remark: Under the Hilbert-Schmidt inner product the Paulis form an orthogonal (not orthonormal) basis for the space of  $2 \times 2$  matrices.

3. (**\*Homework\***) (20 marks) *The Bloch ball.*

The Bloch ball (which is the generalization of the Bloch sphere) is a representation of all qubit states, both pure and mixed. This is the unit ball in  $\mathbb{R}^3$ . The surface of this ball, which is the unit sphere, is called the Bloch sphere.

Consider any state  $\rho$  of a 2-level system (qubit). We will map this to a point in the unit ball. This will be the Bloch ball representation of  $\rho$ .

- (a) Given a qubit state  $\rho$ , show that there exist unique real numbers  $s_i, s_x, s_y, s_z$  such that

$$2\rho = s_i I + s_x X + s_y Y + s_z Z.$$

Hint: Use the property that  $\rho$  is a Hermitian operator.

- (b) Argue that  $s_i = 1$ .

Hint: Use the fact  $\text{Tr}(\rho) = 1$ .

- (c) Compute the eigenvalues of  $\rho$  in terms of  $s_x, s_y, s_z$ . You will be able to show that the two eigenvalues (which are both real) are

$$\frac{1 \pm s}{2}, \quad \text{where } s = \sqrt{s_x^2 + s_y^2 + s_z^2}.$$

- (d) Since  $\rho$  is a density operator what is the allowable range of values for  $s$ ?

Hint:  $\rho$  must be a positive operator. What do we know about its eigenvalues?

- (e) Argue that  $\rho$  is pure when  $s = 1$  and  $\rho$  is mixed when  $s < 1$ .

Remark: The Bloch ball representation of  $\rho$  is the real vector  $(s_x, s_y, s_z)$ . You have now shown that this real vector lies within the unit ball with center at  $(0, 0, 0)$ . The points on the surface ( $s = 1$ ) are the pure states, and the points in the interior are all mixed states.

The state corresponding to  $s = 0$  is called the *completely mixed* or *maximally mixed* state (what is  $\rho$  when  $s = 0$ ?).

Typically when noise affects a pure state qubit ( $s = 1$ ) it transforms the qubit to mixed state ( $s < 1$ ).

4. *Mixed States and Superposition.*

Consider a photon where the quantum information is stored in its polarization state. The classical polarization states of the photon are  $|0\rangle$  and  $|1\rangle$ , denoting horizontal and vertical polarization states, respectively.

- (a) Suppose a horizontally polarized photon is passed through an optical waveguide that transforms pure states as follows:  $|\psi\rangle \rightarrow H|\psi\rangle$ , where

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad (\text{this is known as the Hadamard gate}).$$

What is the state of the photon after passing through the optical waveguide? Is this a pure state or a mixed state? Is the photon in a classical state after passing through the waveguide?

- (b) Suppose a randomized experiment creates a photon in horizontal polarization with probability  $1/4$  and in vertical polarization with probability  $3/4$ . What is the ensemble representation of this photon? What is the state of this photon?
- (c) If the photon considered in part (b) is passed through the optical waveguide in part (a), what is the state of the photon after it passes through the waveguide?

5. *Cauchy-Schwarz and triangle inequalities.*

The Cauchy-Schwarz inequality states that for any non-zero  $|\phi\rangle, |\psi\rangle \in \mathbb{C}^d$  we have

$$|\langle\phi|\psi\rangle| \leq \|\phi\| \cdot \|\psi\|$$

and equality occurs if and only if  $|\phi\rangle = \alpha|\psi\rangle$  for some  $\alpha \in \mathbb{C}$ . The inequality holds even if  $|\phi\rangle$  or  $|\psi\rangle$  is 0, but it is trivial in this case.

Use the Cauchy-Schwarz inequality to prove the triangle inequality:

$$\| |\phi\rangle + |\psi\rangle \| \leq \|\phi\| + \|\psi\|.$$

Hint: Consider  $\| |\phi\rangle + |\psi\rangle \|^2 = (\langle\phi| + \langle\psi|)(|\phi\rangle + |\psi\rangle)$ .

6. **(\*Homework\*)** (5 marks) *Purity of a state.*

Consider any  $d$ -level quantum system. The *purity* of a state  $\rho$  is defined to be  $\text{Tr}(\rho^2)$ . Show that the following are true for any density operator  $\rho$

- (a)  $\text{Tr}(\rho^2) \leq 1$ ,
- (b)  $\text{Tr}(\rho^2) = 1$  if and only if  $\rho$  is a pure state.

Hint: Use the spectral decompositions of  $\rho$  and  $\rho^2$ .

7. **(\*Homework\*)** (5 marks) *Bounds on purity.*

If  $\rho$  is the state of a  $d$ -level system, you saw in the previous question that the purity  $\text{Tr}(\rho^2) \leq 1$ . We will now derive a lower bound on purity using the Cauchy-Schwarz inequality.

- (a) Show that for any state  $\rho$  of a  $d$ -level system we have

$$\text{Tr}(\rho^2) \geq \frac{1}{d}.$$

Hint: You can use Cauchy-Schwarz to prove this, with  $\| |\phi\rangle \| = \sqrt{\text{Tr}(\rho^2)}$  and  $\| |\psi\rangle \| = \sqrt{d}$ . Identify a  $|\phi\rangle$  and a  $|\psi\rangle$  which will help you reduce purity lower bound to a Cauchy-Schwarz inequality.

- (b) Identify the unique state  $\rho$  for which  $\text{Tr}(\rho^2) = 1/d$ , and argue that there is only one state that satisfies this condition.

Remark: This is called the *maximally mixed* or the *completely mixed* state.

8. Suppose  $\rho$  is any state of a  $d$ -level quantum system, and the rank of  $\rho$  is  $r$ . Argue that if  $\rho$  can be prepared using an ensemble of  $n$  pure states then  $n \geq r$ . That is, to prepare a rank- $r$  state using an ensemble we need at least  $r$  pure states in the ensemble.

Hint: Use the rank-sum inequality  $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$ .